

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0007337		
	Version:	8.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	28-Apr-2016
Title:	SM DG57-DG517. Operation qualification			

DG57 / DG517 Service Manual

OPERATION QUALIFICATION

Procedure description
Operation Qualification (OQ) testing is performed to verify that the instrument or instrument modules operate as intended. It should be performed in the following cases: • After IQ procedure. • After a corrective action, executing the part related to the corresponding module. • After a preventive maintenance. <i>Note: Document REGD-0007337_8 replaces document REGD-0007337_v7.</i>

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OPERATION QUALIFICATION (OQ).

Operation qualification REGD-0007337 is related to DG-57 service manual (REGD-0000749) and DG-517 service manual (REGD-0009629).

First of all, ensure that the mains switch, which is located in the back side of the instrument, is turned off. Then plug the mains cable, supplied with the instrument, to the DG-57/ DG-517 first and to the mains later.

	DANGER! Electromagnetic compatibility and electrical safety tests have been performed using the cord supplied by the manufacturer. Using cables different to the one supplied may degrade the performances of the instrument concerning electromagnetic compatibility and safety.
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	NOTE: If the instrument has been stored under different environmental conditions than the operation conditions, wait for at least 1 hour under the operation conditions before turning the instrument on.
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Before turning the instrument on, check that the empty waste bottle is connected and both wash solution bottles are filled with expected solutions and connected. Check also that the card waste container is placed correctly.

Instrument:	<i>DG57 / DG517</i>	Form version:	8.0
Serial number:		Code:	
Installation date:		Qualification Date:	

Customer Site:		Customer Name:	
Department:			
Tel:		Fax:	
		E-Mail:	

Service Engineer Name:			
Department:			
Tel:		Fax:	
		E-Mail:	

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1 Installation checking:

Configuration	Result
PC model	
PC serial number	
PC operating system (OS)	
DianaSystem Software version	
Matrox Image Acquisition Card Model	
Reference image is present (if not present, create one)	
Reagents Pack version	

2 Reader module (Technical Service software)

Procedure	Expected Result	Result	Pass/Fail
Reader camera optics verification			
In <i>Settings</i> tab, click <i>Reader</i> button in <i>Camera optics</i> section and then click on <i>Next</i> button 5 times. Wait until Finish button is available (do not press it). Check current brightness and contrast values and then press <i>Cancel</i>	For Matrox Meteor II board, brightness is between 100 and 128 and contrast is 160 For Matrox Morphis board, brightness is between 103 and 114 and contrast is 166		
Reference image verification			
Open the reference image in the working directory (<i>W1xxxx4.tif</i> , where xxxx is the instrument serial number)	The card used for the reference image is in proper conditions (refer to REGD-0000814)		

Note: Print and attach the reference image (if printer available)

3 WADDiagnostic.

Module	Pass/Fail	Comments
Communications		
Settings		
Initializations		
Motor Movements		
Vaccum/Pressure		
Temperature Regulation		
Tubes identification		
Reagents identification		
Level detection		
Cards transport		
Cards reading		
Centrifugation		
Cent. balancing detector		
Solutions bottles level		
Fluid circuit		
Pipeting		
Door, leds and buzzer		

Note: In case of running WADDiagnostic on DG-517 or DG-57 S/N ≥ 057-0004285 'centrifuge balancing detector' step will not apply (Comments NA).

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Procedure	Expected Result	Result	Pass/Fail
Probe dispensation position verification	Holes in the dispensed card film are centred in Y axis and shifted about 1mm to the left part of the instrument in X axis		

4 Temperature (REGD-0000799 : Checking the adjust of the incubator module)

Procedure	Expected Result	Result	Pass/Fail
Set block 1 at 24°C	24°C ± 3°C		
Set block 2 at 24°C	24°C ± 3°C		
Set block 1 at 37°C	37°C ± 1°C		
Set block 2 at 37°C	37°C ± 1°C		

5 Centrifuge speed and time (REGD- 0000832):

Procedure	Expected Result	Result	Pass/Fail
Set speed at 1228 rpm	1228 ± 6 rpm		
Set time at 540 seconds	540 ± 10 seconds		

6 Reagents spinning (REGD- 0000797):

Procedure	Expected Result	Result	Pass/Fail
Measure slow spinning speed	> 100 rpm		
Measure fast spinning speed	> 400 rpm		
Measure fastest spinning speed	> 500 rpm		
Check that all reagents carrousel positions are able to spin	Positions 1 to 8 and 10 to 17 are able to spin		

7 Real batch test

Procedure	Expected Result	Result	Pass/Fail
User/Operator executes 4 ABO-Rh profiles in Wadiana C. application	Correct results For negative results, the size of the button is homogeneous between wells		

Note: Print and attach the resulting batch listing

Comments:

Service Engineer signature:

Customer signature:

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DOCUMENT CHANGE HISTORY

Version	Date	Change	Rationale
4.0	2011/05/18	Replaced commercial name by project name.	Delete commercial name.
4.0	2011/05/18	Note: Document REGD-0001337_v4 replaces document REGD-0001337_v3.	Document version changed.
4.0	2011/05/18	Added 'procedure description' section.	Unify service manual documents style.
4.0	2011/05/18	Updated 'check computer' point: - Deleted REGD documents reference. -Modified 'Matrox video card installed' section	Unify OQ to DG-57/DG517.
4.0	2011/05/18	Update 'WADDiagnostic' point: - Deleted REGD documents reference. -Deleted 'clot detection' step. -Added 'Note'.	Unify OQ to DG-57/DG517.
4.0	2011/05/18	Header title has changed.	Previous header title has been substituted by project and document title.
5.0	2011/09/20	Added service manual index references.	Document related to service manual.
5.0	2011/09/20	Added qualification date.	Qualification date can be different than installation date.
5.0	2011/09/20	Added number of pages.	Number of pages in the page bottom side.
5.0	2011/09/20	Description simplified.	Procedure description section only describes the procedure.
6.0	October 2012	Updated WADdiagnostic note	'centrifuge balance detector' checking will also not apply to DG-57.
6.0	October 2012	Corrected holes dispensed card film checking.	Ensure correct card dispensed position.
6.0	October 2012	Added reagents spinning verification.	Ensure correct reagents spinning.
7.0	December 2014	Change of structure	Make the OQ procedure structure similar to those of other Diagnostic Grifols projects
7.0	December 2014	Reader module verification	New verifications for reader module
7.0	December 2014	Centrifuge module verification	Speed and time values are those used in routine
8.0	April 2016	Installation checking points	Checking points updated